Parametric Modeling of Serpentine Waveguide Traveling Wave Tubes

Kasra Rouhi, *Graduate Student Member, IEEE*, Robert Marosi, *Member, IEEE*, Tarek Mealy, Alexander Figotin, and Filippo Capolino, *Fellow, IEEE*

***Abstract*— A simple and fast model for numerically calculating small-signal gain in serpentine waveguide traveling-wave tubes (TWTs) is described. In the framework of the Pierce model, we consider 1-D electron flow along a dispersive single-mode slow wave structure (SWS), accounting for the space-charge effect. The analytical model accounts for the frequency-dependent phase velocity and characteristic impedance obtained using various equivalent circuit models from the literature, validated by comparison with full-wave eigenmode simulation. The model includes a relation between the modal characteristic impedance and the interaction (Pierce) impedance of the SWS, including also an extra correction factor that accounts for the variation of the electric field distribution and hence of the interaction impedance over the beam cross section. By applying boundary conditions to our generalized Pierce model, we compute both the theoretical gain of a TWT and all the complex-valued wavenumbers of the hot modes versus frequency and compare our results with computationally intensive particle-in-cell (PIC) simulations; the good agreement in the comparison demonstrates the accuracy and simplicity of our generalized model. For various examples where we vary the average electron beam (e-beam) phase velocity, average e-beam current, number of unit cells, and input radio frequency (RF) power, we demonstrate that our model is robust in the small-signal regime. The purpose of this article is not to design a TWT with performance that competes with previous ones, but to develop an accurate and simple model to predict TWT performance that can be used as a design tool.**

***Index Terms*— Dispersion, electron beam (e-beam) devices, Pierce theory, serpentine waveguide, slow wave structure (SWS), traveling-wave tube (TWT).**

I. Introduction

THE traveling-wave tube (TWT) is a type of common microwave vacuum electron tube that has been widely used for applications such as communication, radar, and electronic countermeasures [1], [2], [3], [4]. Among the different kinds of TWTs, the serpentine waveguide TWT has advantages over other kinds of millimeter wave TWTs (e.g., helix TWT, coupled cavity (CC) TWT, ring-bar TWT) due to its moderate bandwidth with power-handling capacity at higher frequencies and its compatibility with planar fabrication using lithography or micromachining [5], [6], [7], [8], [9]. The slow wave structure (SWS) of the serpentine waveguide TWT is formed by bending rectangular waveguides in the electric field plane (E-plane). Also, a cylindrical electron beam (e-beam) is transported through the cylindrical beam tunnel to interact with the radio frequency (RF) propagating wave. Although the serpentine waveguide SWS's performance is limited by its low interaction impedance and interaction efficiency, many schemes of enhancing the on-axis interaction impedance and also enhancement of interaction efficiency have been proposed [10], [11], [12], [13], [14], [15].

In order to analyze the e-beam and electromagnetic (EM) wave dynamics of a serpentine waveguide TWT, it is necessary to examine the EM characteristics of the SWS. Various analytical models have been developed for its characterization. In 1987, Dohler et al. [5] proposed a simple analytical method for determining the dispersion characteristics and the interaction impedance of the EM modes in the serpentine waveguide. Liu [6] suggested an analytical formulation adding the effect of bends. Then, researchers developed a closed-form algebraic dispersion relation based on an equivalent circuit model that also considered the effect of mismatch between straight and bend sections as well as an approximate model for beam holes [16], [17]. A thorough equivalent circuit analysis of serpentine waveguides by modeling the effect of beam tunnels as orthogonal stubs was developed by Booske et al. [18] for the calculation of dispersion characteristics, following the approach of transmission line (TL) cascading networks and benchmarked using 3-D simulations with Ansys HFSS, MAFIA, and CST Studio Suite. Recently, Antonsen et al. [19] developed a hybrid model consisting of a combination of TL segments and lumped electrical elements, which is utilized to analyze serpentine waveguide dispersion characteristics and interaction impedance. The model also captures the effects of asymmetric fields and beam tunnel misalignment. Although some commercial full-wave simulation software like Ansys HFSS and CST Studio Suite are versatile and can analyze SWS characteristics, simulation times are longer than analytical methods. Therefore, analytical methods are preferred for quickly iterating through and optimizing various SWS designs.

Manuscript received 11 August 2023; revised 30 November 2023 and 8 March 2024; accepted 12 March 2024. Date of publication 11 April 2024; date of current version 21 June 2024. This work was supported by the Air Force Office of Scientific Research (AFOSR) Multidisciplinary Research Program of the University Research Initiative (MURI) administered through The University of New Mexico under Grant FA9550-20-1-0409. The review of this article was arranged by Senior Editor M. Blank. *(Corresponding author: Filippo Capolino.)*

Kasra Rouhi, Robert Marosi, Tarek Mealy, and Filippo Capolino are with the Department of Electrical Engineering and Computer Science, University of California at Irvine, Irvine, CA 92697 USA (e-mail: kasra.rouhi@uci.edu; rmarosi@uci.edu; tmealy@uci.edu; f.capolino@uci.edu).

Alexander Figotin is with the Department of Mathematics, University of California at Irvine, Irvine, CA 92697 USA (e-mail: afigotin@uci.edu).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TPS.2024.3379944.

Digital Object Identifier 10.1109/TPS.2024.3379944

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